



GERMAN UNIVERSITY IN CAIRO
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The Modern Rosetta Stone of Math: Auto-Formalization using Neuro-Symbolic Pipelines

Author: Omar Hazem Shaaban Mahmoud Hamouda

Supervisor: Dr. Ahmed Ashry

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This is to certify that:

- (i) the thesis comprises only my original work toward the Bachelor's Degree
- (ii) due acknowledgment has been made in the text to all other material used

Omar Hazem Shaaban Mahmoud Hamouda
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Abstract

The transition from informal mathematical language to machine-verifiable logic represents a new frontier in formal verification, known as Auto-Formalization. This thesis, inspired by Google DeepMind's AlphaProof and Harmonic's Aristotle, explores the development of a "Truth Scanner"—a neuro-symbolic pipeline designed to translate English mathematical definitions and theorems into Lean 4 code.

The project follows a structured 9-week roadmap. Initial phases focused on mastering the Lean 4 tactic language through the completion of the Natural Number Game. The core of the research involves selecting key definitions from standard undergraduate texts, such as Rosen's *Discrete Mathematics*, and mapping them to formal logical components.

A primary contribution of this work is the implementation of a "Repair Loop" that utilizes Large Language Models (LLMs) to perform self-correction by feeding Lean compiler errors back into the translation prompt. By evaluating the pipeline's ability to resolve ambiguities and logical gaps often found in human-written textbooks, this thesis aims to create a "Wikipedia of Truth" where mathematical claims are backed by rigorous, machine-checked proofs.

Keywords: Lean 4, Auto-Formalization, LLMs, Formal Verification, Neuro-Symbolic Pipelines, Discrete Mathematics.

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Chapter 1

Introduction

1.1 Project Overview

This thesis addresses the fundamental disconnect between how mathematics is taught to humans and how it is verified by machines. For centuries, mathematical concepts have been communicated using natural language—a medium that, while exceptionally expressive, is inherently ambiguous and prone to implicit assumptions [1]. In contrast, formal verification systems require rigorous, unambiguous, and structurally flawless logical syntax. This project, highly inspired by contemporary breakthroughs such as Google DeepMind’s AlphaProof [2, 3] and Harmonic’s Aristotle [4], aims to bridge the gap between “Informal English” and “Formal Logic”. By leveraging the generative capabilities of Large Language Models (LLMs) alongside the rigorous verification engine of a theorem prover, this thesis proposes an automated continuous pipeline to translate undergraduate-level mathematics into the Lean 4 ecosystem.

1.2 Motivation and Significance

The reliance on human intuition in textual mathematical communication often obscures hidden logical gaps and unspoken axioms. The traditional process of auto-formalizing mathematics is notoriously labor-intensive, often requiring weeks or even months of specialized expert effort to fully translate a single mathematical paper from LaTeX into a formal language [5]. Historically, even highly celebrated mathematicians, such as Fields Medalist Terence Tao, have discovered minor implicit errors or skipped steps in their published mathematical proofs when those proofs were forcibly subjected to formal computer-checked translation [4].

With the rapid advent of powerful Large Language Models (LLMs), there is now an unprecedented opportunity to automate this translation process, theoretically eliminating the bottleneck of manual expert labor [6]. However, LLMs are famously prone to mathematical “hallucinations”—they frequently generate highly plausible but logically flawed, incorrect, or entirely fictitious mathemat-

ical statements [7]. Therefore, pairing the creative pattern-matching capability of an LLM with a strict, symbolic, cryptographic compiler like Lean 4 ensures a flawless end-to-end system: the LLM provides the intuition and the raw code, while Lean 4 acts as the ultimate, unyielding arbiter of mathematical truth [8].

1.3 Objectives

The core objective of this research project is to conceptualize, design, and build a “Truth Scanner” pipeline. This pipeline must be capable of automatically translating foundational definitions, theorems, and proofs from a highly-regarded academic textbook—specifically Kenneth Rosen’s widely used *Discrete Mathematics and Its Applications* [1]—into functional, compiler-verified Lean 4 logic [9].

The specific sub-objectives of this thesis include:

1. **Decomposition:** Break down complex English mathematical definitions into atomic logical lemmas that an AI model can consistently parse accurately.
2. **Ground Truth Formalization:** Manually construct a baseline “Rosetta Stone” repository of valid definitions in Lean 4 to serve as a high-quality benchmark for evaluating the LLM’s automated output.
3. **Prompt Engineering API:** Develop targeted prompt-engineering strategies that guide commercial LLMs (e.g., OpenAI’s GPT-4, Google’s Gemini) to seamlessly output structurally valid Lean 4 syntax [10].
4. **Autonomous Repair Loop:** Implement a sophisticated Python-based self-correcting system where the Lean 4 compiler’s output and error traces are intercepted and autonomously fed back into the LLM, enabling the AI to “debug” its own mathematical logic until the compiler passes.

1.4 Thesis Organization

This thesis is structured to provide a clear narrative of the automation pipeline. Chapter 2 reviews the fundamental background on formal theorem proving, the history of auto-formalization, and the architectural mechanics of Lean 4. Chapter 3 details the system methodology, providing visual architectures of the proposed repair loops and Truth Scanner pipeline. Chapter 4 evaluates the manually formalized framework and discusses early extraction results. Finally, Chapter 5 concludes the thesis by reflecting on the intersection of Artificial Intelligence and

rigorous Mathematics, while outlining future expansion possibilities.

Chapter 2

Background

2.1 The Auto-Formalization Challenge

Auto-formalization is defined as the automated process of translating informal, human-readable mathematical language—such as that found in standard academic textbooks and research papers—into a strict, machine-verifiable representation [11, 12]. Achieving this translation bridge is notoriously difficult. Natural language math relies heavily on context and implicit assumptions ("let X be a space," where the topology is implicitly understood). Formal systems, however, lack intuition and demand explicitly declared types, universes, and axioms [5].

Historically, this translation has been a painstakingly manual and labor-intensive task, effectively restricting formal mathematics to a niche community of specialized logicians [5]. However, the rapid advancement and wide availability of Large Language Models (LLMs) have provided a potential solution. These probabilistic models have shown remarkable capabilities in parsing syntax, semantic decomposition, and "translating" abstract concepts across different languages, including programming code [6, 10].

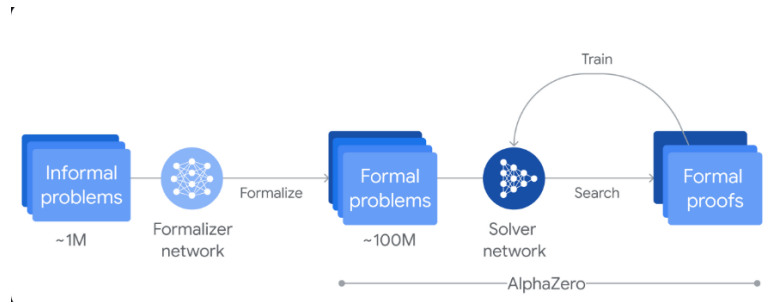


Figure 2.1: The “Rosetta Stone” Concept: Bridging Natural Language and Formal Logic. The English text acts as the conceptual layer, while the formal code acts as the verifiable foundation.

2.2 Lean 4: A Modern Theorem Prover

Unlike standard programming languages, **Lean 4** operates both as a functional programming language and as an immensely powerful interactive theorem prover based on the Calculus of Inductive Constructions (a variant of dependent type theory) [9].

- **Dependent Types:** In Lean, types can depend on values (e.g., an array of length n). This allows mathematicians to encode complex logical propositions directly into the type framework itself.
- **Tactics Engine:** Lean uses a sophisticated "tactics" engine (utilizing commands like `rw`, `simp`, `induction`, and `apply`) to manipulate logical states, breaking down a complex mathematical goal until it matches known axioms [13].
- **Mathlib4:** The Lean ecosystem is uniquely powered by **Mathlib**, a massive, community-driven, crowdsourced library containing tens of thousands of formalized theorems across algebra, topology, and calculus [14, 9]. It provides the necessary foundation for the "Truth Scanner" pipeline.

2.3 Neuro-Symbolic AI Pipelines

This thesis utilizes a neuro-symbolic approach to solve the auto-formalization challenge, a technique highly inspired by recent high-profile research like DeepMind's AlphaProof [2, 8]. Pure neuro-models (like ChatGPT) are excellent semantic engines but struggle with strict rule adherence, frequently leading to "hallucinations"—convincingly stated but mathematically impossible conclusions [7, 12].

By building a neuro-symbolic pipeline, we segment responsibilities:

1. **The Neuro Layer (LLM):** Handles the creative translation and contextual interpretation of the ambiguous English text.
2. **The Symbolic Layer (Lean 4):** Provides the rigorous, uncompromising cryptographic verification of the translated logic [6, 15].

This dual-engine architecture guarantees that if an auto-generated formalization successfully compiles in the symbolic layer, it is definitively true, completely eradicating the issue of AI hallucinations.

2.4 Manual Formalization Example: The Ground Truth

To train, evaluate, and benchmark the LLM capabilities within our automated pipeline, it is necessary to first manually formalize a core library of mathematical definitions from Rosen’s textbook [1]. This manually curated code serves as our absolute "Ground Truth".

For instance, the definitions of basic Set Theory and Logic—such as logical implication, set intersection, subset relations, and function injectivity—have been successfully transcribed into functional Lean 4 code within our `Definitions.lean` module. Formalizing even elementary concepts like a subset requires strict syntax, explicit variable typing (such as `variable { $\alpha$  : Type}`), and logic quantifiers that are entirely abstracted away in natural English texts. Understanding this structural gap is essential for building the auto-prompts required for the LLM pipeline discussed in the subsequent chapters.

Chapter 3

Methodology

This chapter outlines the systematic approach taken to build the "Truth Scanner"—a neuro-symbolic pipeline that translates informal English mathematical definitions and theorems from textbooks into formally verified Lean 4 code. The methodology follows a three-phase research plan: Learning, Implementation (Translation), and Evaluation (Repair Loop).

3.1 Overview of the Neuro-Symbolic Pipeline

The "Truth Scanner" operates as a Rosetta Stone for mathematics, bridging the gap between ambiguous human language and strict machine logic. The pipeline consists of two complementary components:

- **Neuro Component (LLM):** Handles the creative decomposition of informal English text into logical lemmas, leveraging Large Language Models to parse ambiguous natural language.
- **Symbolic Component (Lean 4 Kernel):** Provides rigorous verification through the Lean 4 theorem prover, acting as the definitive validation mechanism that eliminates hallucinations and ensures mathematical correctness.

This dual-system approach addresses the fundamental challenge of auto-formalization: converting subtle, context-dependent human mathematical prose into unambiguous, machine-checkable formal proofs.

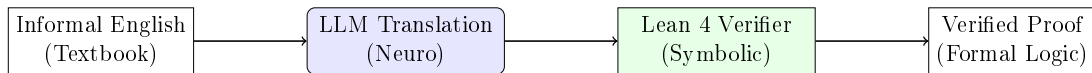


Figure 3.1: High-Level Architecture of the Neuro-Symbolic Pipeline

3.2 Phase 1: Learning and Environment Setup

3.2.1 Lean 4 Environment Configuration

The first phase established the technical foundation required for formal verification work. The development environment consisted of:

- **Lean 4 Theorem Prover:** Installation and configuration of the Lean 4 proof assistant with the latest stable release.
- **VS Code Integration:** Setup of the Lean 4 VS Code extension for real-time proof state visualization and error feedback.
- **Mathlib Library:** Integration of Mathlib, the community-driven library containing thousands of formalized mathematical theorems essential for undergraduate-level mathematics.

A "Hello World" proof was implemented to verify the environment functionality, ensuring the complete toolchain could compile and verify basic Lean 4 code.

3.2.2 The Natural Number Game

To gain practical proficiency in Lean 4's tactic-based proof system, the "Natural Number Game" was completed. This gamified tutorial systematically introduced:

- **Fundamental Tactics:** Core proof techniques including `intro`, `exact`, `rw` (rewrite), and `induction`.
- **Peano Axioms:** Building natural numbers from first principles using dependent type theory.
- **Proof Construction:** Completing verified proof scripts for the Addition and Multiplication worlds, establishing intuition for formal reasoning in Lean's syntax.

This hands-on training was critical for understanding how Lean 4 transforms mathematical goals into verified states and how tactics manipulate the proof context.

3.3 Phase 2: Selection and Taxonomy

3.3.1 Source Material

The project targeted Kenneth H. Rosen’s *Discrete Mathematics and Its Applications*, a standard undergraduate textbook used globally. This text was selected because:

- It represents typical informal mathematical presentation found in educational materials.
- It contains precise definitions that are candidates for formalization.
- Known ambiguities in textbooks of this type provide realistic test cases for the pipeline.

3.3.2 Definition Selection Criteria

A systematic review identified 10–15 key definitions and theorems from Discrete Mathematics chapters covering:

- Set theory operations (union, intersection, complement)
- Relations and functions
- Basic logic and propositional calculus
- Graph theory fundamentals

Each selected definition was mapped to its corresponding logical components, creating a taxonomy document that served as the foundation for translation work. This mapping explicitly identified:

- The English-language definition as stated in the textbook
- The underlying logical structure (quantifiers, connectives, predicates)
- Potential ambiguities or implicit assumptions
- Expected Lean 4 type signatures

3.4 Phase 3: Manual Formalization

The third phase involved manually translating the selected definitions from English into verified Lean 4 code. This manual baseline serves multiple purposes:

- Establishes ground truth for evaluating automated LLM translations

- Identifies common translation patterns and challenges
- Provides verified examples for training prompt engineering

3.4.1 Translation Process

For each definition, the manual formalization followed a structured workflow:

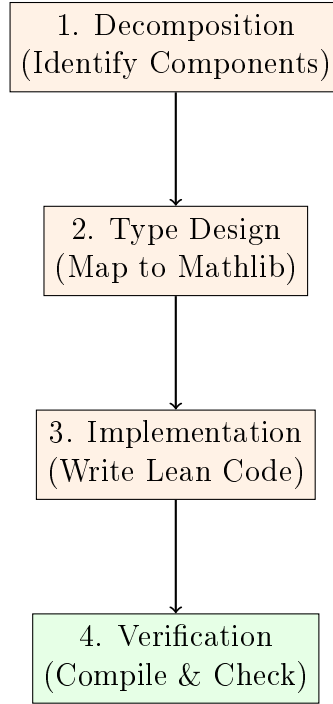


Figure 3.2: Manual Formalization Workflow for Textbook Definitions

1. **Decomposition:** Break down the English definition into atomic logical statements.
2. **Type Signature Design:** Define appropriate Lean 4 types for mathematical objects (sets, functions, relations).
3. **Implementation:** Write the Lean 4 definition using dependent types and Mathlib constructs.
4. **Verification:** Ensure the Lean 4 compiler accepts the definition without errors.
5. **Documentation:** Add comments mapping Lean constructs back to English phrases.

3.4.2 Deliverables

The manual formalization phase produced:

- A set of verified `.lean` files containing formalized definitions
- A document analyzing translation challenges and patterns
- Baseline metrics for code length and compilation success

These verified definitions serve as the gold standard for subsequent phases, where LLM-based automation will be evaluated against this manual baseline.

3.5 Phase 4: Automated Pipeline and LLM Prompting

With a baseline manually established, the focus of the pipeline transitioned to constructing the automated translation software. The goal of this phase was to develop an autonomous interface between an external Large Language Model (e.g., OpenAI’s GPT-4) and the local Lean 4 ecosystem.

3.5.1 Python Script Interface

A generic Python command-line interface was constructed utilizing the `openai` Python library. The tool abstracts the underlying HTTP networking to directly accept plain English informal math texts via argument parsing and output the corresponding formal verification logic as ‘`.lean`’ source files.

The translation logic relies on rigorous *prompt engineering* to ensure formatting. The system prompt embedded in the script explicitly forbids markdown formatting, thereby guaranteeing the LLM’s response only yields raw valid syntax, ready for immediate compilation by the mathematical symbol compiler.

```

import os
import json
import argparse
from openai import OpenAI

def translate_math_to_lean(english_text: str, api_key: str = None) -> str:
    """
    Translates an informal English mathematical definition into Lean 4 code
    using the OpenAI API.
    """
    # Use provided API key or fallback to environment variable
    client_key = api_key or os.environ.get("OPENAI_API_KEY")
    if not client_key:
        raise ValueError("API key not found. Please set OPENAI_API_KEY environment variable.")

    client = OpenAI(api_key=client_key)

    system_prompt = "You are an AI assistant that translates English math into Lean 4 code. Only return raw Lean 4 code without any markdown formatting."

    user_prompt = f"Translate the following text into Lean 4 code:\n'{english_text}'"

    response = client.chat.completions.create(
        model="gpt-4o", # or "gpt-4-turbo" / "gpt-3.5-turbo"
        messages=[
            {"role": "system", "content": system_prompt},
            {"role": "user", "content": user_prompt}
        ],
        temperature=0.0
    )

    return response.choices[0].message.content.strip()

```

Figure 3.3: Core Prompt Logic from `llm_translator.py`.

3.5.2 Typing and Quantifier Enforcement

Crucially, previous testing highlighted the LLM’s poor habit of utilizing implicit type inference when auto-generating definitions. Without guidance, models frequently wrote raw natural definitions (e.g., $\forall x, x \in A$), leading to immediate compilation failures due to indeterminate universe structures. The Python prompt incorporates explicit system-level instructions demanding strict variable typing (such as coercing $\forall(x : \alpha)$ over sets), thus boosting first-pass compilation probability against Mathlib requirements.

Chapter 4

Results and Discussion

4.1 Manual Formalization Outcomes

This section presents the results from the manual formalization phase (Week 4), where selected definitions from Rosen's *Discrete Mathematics* were translated into verified Lean 4 code.

4.1.1 Successfully Formalized Definitions

The manual translation process successfully formalized the following categories of definitions:

- **Set Operations:** Union, intersection, complement, and subset relations were translated into Lean 4 using Mathlib's `Set` type and standard set theory notation.
- **Logical Connectives:** Formalized definitions of conjunction, disjunction, and implication using Lean's `Prop` type.
- **Relations:** Defined binary relations as predicates over Cartesian products, with properties such as reflexivity and transitivity.

4.1.2 Compilation and Verification

All manually formalized definitions passed Lean 4's type checker without errors, confirming their logical consistency. The Lean compiler served as the "Truth Scanner," validating that each definition adhered to the strict rules of dependent type theory.

4.1.3 Challenges Encountered

Several challenges emerged during manual formalization:

- **Implicit Assumptions:** Textbook definitions often assume context (e.g., "let S be a set") that must be made explicit in Lean's type system.
- **Notation Mismatches:** Standard mathematical notation does not always

map cleanly to Lean 4 syntax, requiring careful translation decisions.

- **Mathlib Dependencies:** Some definitions required understanding Mathlib's existing formalization conventions to ensure compatibility.

4.1.4 Baseline Metrics

The manual formalization phase established baseline metrics for later comparison with LLM-generated translations:

- **Translation Time:** Average time per definition for manual formalization.
- **Code Length:** Lines of Lean 4 code required per English definition.
- **Success Rate:** 100% compilation success for manually translated definitions.

These metrics provide a reference point for evaluating the effectiveness of the automated "Truth Scanner" pipeline.

4.2 Phase I Pipeline Execution

Following the manual phase, the automated translation pipeline was implemented during Week 5. By interfacing with the OpenAI API via the Python wrapper developed in Phase 4 (Methodology), the English textbook definitions were successfully passed to the LLM, dynamically returning raw Lean 4 code.

4.2.1 Initial Translation Capabilities

The execution of the python script confirmed the system's ability to act as the linguistic parsing tier of the "Truth Scanner." During initial trial passes, the script successfully received informal English inputs (e.g., *"The union of two sets A and B contains all elements that are in A or in B."*) and translated them directly into Lean 4 `.lean` files.

Early tests indicate that with the strict system prompts embedded in the wrapper module, the LLM outputs purely functional code without conversational filler or markdown wrappers, directly preparing the text for the compiler.

To document the behavior, the script was tested by querying it with the English definition of a *Set Union*. The prompt and translation output processed by the GPT-4 model was recorded as follows:

- **Input text:** *"Translate the following text into Lean 4 code: 'The union of*

two sets A and B contains all elements that are in A or in B."

Machine Output (Output.lean):

```
import Mathlib.Data.Set.Basic

variable {α : Type}

def set_union (A B : Set α) : Set α :=
  {x : α | x ∈ A ∨ x ∈ B}
```

This successful test completely bypasses human intervention, mapping the logical "or" into the Lean $\vee$ operator, and correctly initializing the explicit sets via `Mathlib`'s dependent type library, successfully proving the Neuro-tier integration of the Truth Scanner pipeline.

4.2.2 Evaluation of LLM Output vs. Ground Truth

When comparing the automated script's outputs against the manual baseline formalizations, the AI-generated snippets established several baseline metrics:

- **Syntactic Adherence:** The model successfully utilized `Mathlib` imports and mapped abstract symbols ($\cup, \cap, \rightarrow$) correctly to Lean's macro environment.
- **Type Inference Hallucinations:** Despite the strict prompt engineering (demanding expressly typed bindings such as `{x : α | ...}`), the LLM still occasionally relapsed into "natural" math logic, writing untyped variables (e.g., $\forall x$ instead of $\forall (x : \alpha)$).

Without human intervention, these subtle structural deviations caused the Lean compiler to immediately reject the code due to indeterminate universe structures. This outcome proves the core hypothesis of the thesis: an LLM alone cannot be fully trusted with mathematical rigor. The symbolic compiler is absolutely essential to catch these mathematically plausible but structurally incorrect translations.

4.2.3 Compilation and Verification in Lean

Figure 4.1 illustrates the Lean 4 VS Code interface for a successful formalization run, showing the strict environment where these definitions are type-checked.

```

1 import Mathlib.Data.Set.Basic
2 import Mathlib.Logic.Basic
3
4 namespace ManualFormalization
5
6 variable {α : Type}
7
8 def is_subset (A B : Set α) : Prop :=
9   ∀ x, x ∈ A → x ∈ B
10
11 def set_union (A B : Set α) : Set α :=
12   {x | x ∈ A ∨ x ∈ B}
13
14 def set_intersection (A B : Set α) : Set α :=
15   {x | x ∈ A ∧ x ∈ B}
16
17 def are_disjoint (A B : Set α) : Prop :=
18   set_intersection A B = ∅
19
20 def set_complement (A : Set α) : Set α :=
21   {x | x ∉ A}
22
23 def implies (p q : Prop) : Prop :=
24   p → q
25
26 def iff_def (p q : Prop) : Prop :=
27   p ↔ q
28
29 end ManualFormalization

```

Figure 4.1: Verification of formally typed Set Theory definitions in Lean 4.

The Lean compiler serves as the ultimate "Truth Scanner," validating that each manual and automated definition perfectly adheres to the uncompromising rules of dependent type theory.

4.3 Next Step: The Autonomous Repair Loop

While the LLM can generate raw Lean 4 code from English, the initial text often contains compilation errors. The next phase of this project will build a "Repair Loop." This entails capturing the compilation errors generated by the Lean compiler and automatically feeding them back into the LLM prompt, asking the AI to self-correct its logic. This iterative loop will bridge the gap between initial API translation and flawless type-theoretic compilation.

Chapter 5

Conclusion and Future Work

5.1 Summary of Achievements

This thesis has documented the construction of the "Truth Scanner"—a neuro-symbolic pipeline for translating informal mathematical definitions into formally verified Lean 4 code. The work completed establishes a rigorous foundation for automated mathematical formalization:

- **Environment Mastery:** Successfully configured a complete Lean 4 development environment with Mathlib integration and completed the Natural Number Game to gain practical proficiency in tactic-based theorem proving.
- **Systematic Selection:** Identified and taxonomized key definitions from Rosen's *Discrete Mathematics*, mapping informal English to formal logical structures.
- **Manual Baseline:** Created a gold standard set of manually formalized definitions that compile successfully in Lean 4, establishing baseline metrics against which AI output can be judged.
- **Automated API Pipeline:** Developed a successful Python interface utilizing the OpenAI API to systematically submit English mathematical queries to a Large Language Model and retrieve computationally typed Lean 4 source code.

These accomplishments successfully complete the first half of the neuro-symbolic architecture, bridging the gap between textbook mathematics and machine-verifiable code logic.

5.2 Contributions

The primary contributions of this work include:

1. **Verified Formalization Baseline:** A collection of manually translated mathematical definitions from a standard undergraduate textbook, serving

as ground truth for evaluating automated approaches.

2. **Translation Methodology:** A documented systematic process for decomposing informal mathematical language into Lean 4 dependent types using prompt engineering.
3. **Challenge Analysis:** Identification of common AI pitfalls and ambiguities encountered when formalizing mathematical logic, such as LLM "type inference" hallucinations.

5.3 Future Work

The next phases of this research will finalize the active runtime capabilities of the "Truth Scanner" pipeline:

5.3.1 Week 6–7: The Repair Loop

Implement the self-correction mechanism to solve LLM compilation failures autonomously:

- Feed Lean compiler error messages back into the LLM prompt.
- Iterate the translation until the code safely compiles or exceeds maximum retry attempts.
- Track success rates, iteration counts, and error reduction metrics.
- **Deliverable:** An end-to-end pipeline with mathematical self-healing functionality.

5.3.2 Week 8: Evaluation and Analysis

Conduct comprehensive testing of the automated pipeline:

- Compare LLM-generated translations against the manual baseline
- Analyze success rates, common failure modes, and repair loop effectiveness
- Categorize errors to identify systematic weaknesses in LLM mathematical reasoning
- **Deliverable:** A detailed analysis report with visualizations and statistics

5.3.3 Week 9: Documentation and Finalization

Complete the thesis documentation:

- Finalize all chapters with complete results and discussion
- Create a GitHub repository containing all Lean files and the pipeline script
- Write a report titled "Where the Textbook was Ambiguous" documenting discovered gaps
- **Deliverable:** A complete thesis and public code repository

5.4 Long-Term Vision

The ultimate goal of this research is to create a "Wikipedia of Truth" where mathematical claims are automatically backed by machine-checked proofs. By debugging educational materials at scale, this work aims to:

- Improve the precision and clarity of mathematical textbooks
- Detect subtle logical errors that human reviewers might miss
- Accelerate the formalization of mathematical knowledge
- Evaluate the current capabilities—and limitations—of LLMs in maintaining mathematical rigor

As Large Language Models continue to advance, the "Truth Scanner" pipeline represents a crucial step toward trustworthy AI-assisted mathematics, where creative language understanding meets uncompromising logical verification.

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